IGCSE 0607 Practice Questions — E8.3 — Combined Questions & Markscheme

Magnitude of a vector

Total Questions: 9 — **Total Marks:** 57 Marks

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  → **Paper 2 (Non-Calculator)** *(5 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*

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1 Vector $\mathbf{v}$ is given as

$$\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$$

Work out $|\mathbf{v}|$, the magnitude of vector $\mathbf{v}$.

..... **[2]**

2 Given the vectors $\mathbf{p} = \begin{pmatrix} -8 \\ 15 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$.

(a) Calculate $|\mathbf{p}|$.

..... **[2]**

(b) Calculate $|\mathbf{p} + 2\mathbf{q}|$.

..... **[3]**

3 Vector $\mathbf{u} = \begin{pmatrix} x \\ x+2 \end{pmatrix}$.

(a) The magnitude of vector **u** is 10.
Show that $x^2 + 2x - 48 = 0$.

[3]

(b) Solve the equation

$$x^2 + 2x - 48 = 0$$

$x =$ ..... , $x =$ ..... **[2]**

(c) Hence write down the two possible column vectors for **u**.

[2]

4 Points A, B, and C have coordinates $(3, -2)$, $(-5, y)$, and $(7, 1)$ respectively.

(a) Calculate $|\vec{AC}|$.

..... **[2]**

(b) The magnitude of vector $\vec{AB}$ is 17.
Work out the two possible values of y.

$y =$ ..... , $y =$ ..... **[5]**

5 **(a)** The points A and B have coordinates $(2, -3)$ and $(9, 4)$ respectively.
Work out the column vector $\vec{AB}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Calculate $|\vec{AB}|$.
Give your answer correct to 3 significant figures.

..... **[2]**

(c) Vector $\mathbf{p} = \begin{pmatrix} k \\ 4 \end{pmatrix}$.
Given that $|\mathbf{p}| = 5$, find the two possible values of k.

$k =$ ..... , $k =$ ..... **[3]**

6 Given the vectors $\mathbf{u} = \begin{pmatrix} 2k \\ k-3 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} k+1 \\ 2 \end{pmatrix}$.

(a) Given that $|\mathbf{u} + \mathbf{v}| = \sqrt{50}$, show that $5k^2 + 2k - 24 = 0$.

[4]

(b) Solve the equation

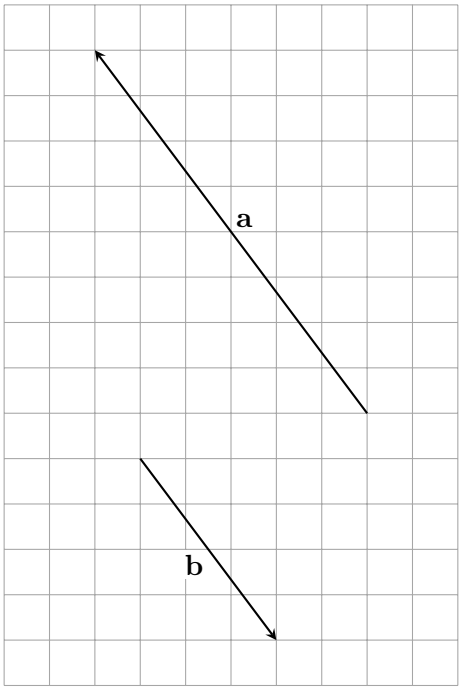
$$5k^2 + 2k - 24 = 0$$

$k =$ ..... , $k =$ ..... **[4]**

(c) Hence, find the two possible column vectors for $\mathbf{u}$.

$\mathbf{u} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ ..... , $\mathbf{u} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ ..... **[2]**

7 The diagram shows vectors **a** and **b** drawn on a square grid.



(a) Work out $|\mathbf{a}|$.

$|\mathbf{a}| =$ ..... **[2]**

(b) Work out $|2\mathbf{a} + 3\mathbf{b}|$.

$|2\mathbf{a} + 3\mathbf{b}| =$ ..... **[3]**

8 The points P, Q and R have coordinates $(2, -3)$, $(k, 5)$ and $(4, 1)$ respectively.

(a) Given that $|\vec{PQ}| = 10$, find the two possible values of k.

$k =$ ..... , or $k =$ ..... **[4]**

(b) Using the positive value of k from part (a), calculate the magnitude of $\vec{QR}$.
Give your answer correct to 3 significant figures.

$|\vec{QR}| =$ ..... **[3]**

9 **(a)** The vector $\mathbf{a} = \begin{pmatrix} p \\ p \end{pmatrix}$, where $p > 0$.

Given that $|\mathbf{a}| = \sqrt{18}$, find the value of p.

$p =$ ..... **[2]**

(b) The vectors $\mathbf{u}$ and $\mathbf{v}$ are given by

$$\mathbf{u} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \qquad \mathbf{v} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

A vector $\mathbf{w}$ is defined as $\mathbf{w} = \mathbf{u} + t\mathbf{v}$.

Given that $|\mathbf{w}| = \sqrt{29}$, find the two possible values of t.

$t =$ ..... , or $t =$ ..... **[5]**

1 $\sqrt{5^2 + (-12)^2}$ or $\sqrt{25 + 144}$ **[M1]**

13 **[A1]**

[Back to Question] *[Total Marks: 2]*

2 **(a)** $\sqrt{(-8)^2 + 15^2}$ or $\sqrt{64 + 225}$ **[M1]**

17 **[A1]**

(b) $\mathbf{p} + 2\mathbf{q} = \begin{pmatrix} -8 \\ 15 \end{pmatrix} + \begin{pmatrix} 4 \\ -8 \end{pmatrix} = \begin{pmatrix} -4 \\ 7 \end{pmatrix}$ **[M1]**

$\sqrt{(-4)^2 + 7^2}$ or $\sqrt{16 + 49}$ **[M1]**

$\sqrt{65}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

3 **(a)** $\sqrt{x^2 + (x+2)^2} = 10$ or $x^2 + (x+2)^2 = 10^2$ **[M1]**

$x^2 + x^2 + 4x + 4 = 100$ **[M1]**

$2x^2 + 4x - 96 = 0$ leading to $x^2 + 2x - 48 = 0$ with no errors seen **[A1]**

(b) $(x+8)(x-6) = 0$ or correct substitution into quadratic formula **[M1]**

$x = -8, x = 6$ **[A1]**

(c) $\mathbf{u} = \begin{pmatrix} -8 \\ -6 \end{pmatrix}$ **[B1]**

$\mathbf{u} = \begin{pmatrix} 6 \\ 8 \end{pmatrix}$ **[B1]**

[Back to Question] *[Total Marks: 7]*

4 **(a)** $\vec{AC} = \begin{pmatrix} 7 - 3 \\ 1 - (-2) \end{pmatrix}$ or $\begin{pmatrix} 4 \\ 3 \end{pmatrix}$ **[M1]**

5 **[A1]**

(b) $\vec{AB} = \begin{pmatrix} -8 \\ y + 2 \end{pmatrix}$ **[B1]**

$\sqrt{(-8)^2 + (y+2)^2} = 17$ **[M1]**

$64 + (y+2)^2 = 289$ or $(y+2)^2 = 225$ **[M1]**

$y + 2 = 15$ or $y + 2 = -15$ **[M1]**

$y = 13$, $y = -17$ **[A1]**

[Back to Question] *[Total Marks: 7]*

5 **(a)** $\begin{pmatrix} 9 - 2 \\ 4 - (-3) \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 7 \\ 7 \end{pmatrix}$ **[A1]**

(b) $\sqrt{7^2 + 7^2}$ or $\sqrt{98}$ **[M1]**

9.90 **[A1]**

(c) $\sqrt{k^2 + 4^2} = 5$ or $k^2 + 4^2 = 5^2$ **[M1]**

$k^2 = 25 - 16$ or $k^2 = 9$ **[M1]**

$k = 3$, $k = -3$ **[A1]**

[Back to Question] *[Total Marks: 7]*

6 **(a)** $\mathbf{u} + \mathbf{v} = \begin{pmatrix} 3k + 1 \\ k - 1 \end{pmatrix}$ **[M1]**

$(3k + 1)^2 + (k - 1)^2 = (\sqrt{50})^2$ **[M1]**

$9k^2 + 6k + 1 + k^2 - 2k + 1 = 50$ **[M1]**

$10k^2 + 4k + 2 = 50 \implies 5k^2 + 2k - 24 = 0$ **[A1]**

(b) $(5k + a)(k + b) = 0$ where $ab = -24$ or $5b + a = 2$, or correct substitution into quadratic formula **[M1]**

$(5k + 12)(k - 2) = 0$ **[M1]**

$k = 2$ **[A1]**

$k = -2.4$ or $-\dfrac{12}{5}$ **[A1]**

(c) Substituting $k = 2$ and $k = -2.4$ into $\begin{pmatrix} 2k \\ k - 3 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 4 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -4.8 \\ -5.4 \end{pmatrix}$ **[A1]**

[Back to Question] *[Total Marks: 10]*

7 **(a)** Determines column vector $\mathbf{a} = \begin{pmatrix} -6 \\ 8 \end{pmatrix}$ **[M1]**

10 **[A1]**

(b) Determines column vector $\mathbf{b} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$ **[B1]**

Calculates resultant vector $2\begin{pmatrix} -6 \\ 8 \end{pmatrix} + 3\begin{pmatrix} 3 \\ -4 \end{pmatrix} = \begin{pmatrix} -12 \\ 16 \end{pmatrix} + \begin{pmatrix} 9 \\ -12 \end{pmatrix} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$ **[M1]**

5 **[A1]**

[Back to Question] *[Total Marks: 5]*

8 **(a)** Expresses vector $\vec{PQ}$ as $\begin{pmatrix} k - 2 \\ 5 - (-3) \end{pmatrix} = \begin{pmatrix} k - 2 \\ 8 \end{pmatrix}$ **[M1]**

Sets up magnitude equation: $(k - 2)^2 + 8^2 = 10^2$ **[M1]**

Simplifies to solve for k: $(k-2)^2 = 36$ **[M1]**

$k = 8$, or $k = -4$ **[A1]**

(b) Finds $\vec{QR}$ using $Q(8,5)$ and $R(4,1)$: $\vec{QR} = \begin{pmatrix} 4-8 \\ 1-5 \end{pmatrix} = \begin{pmatrix} -4 \\ -4 \end{pmatrix}$ **[M1]**

Calculates magnitude: $\sqrt{(-4)^2 + (-4)^2} = \sqrt{32}$ **[M1]**

5.66 **[A1]**

[Back to Question] *[Total Marks: 7]*

9 **(a)** Sets up equation: $p^2 + p^2 = 18$ **[M1]**

3 **[A1]**

(b) Finds expression for $\mathbf{w}$: $\begin{pmatrix} 4-t \\ 1+2t \end{pmatrix}$ **[M1]**

Forms magnitude squared equation: $(4-t)^2 + (1+2t)^2 = 29$ **[M1]**

Expands brackets correctly: $16 - 8t + t^2 + 1 + 4t + 4t^2 = 29$ **[M1]**

Rearranges to a 3-term quadratic: $5t^2 - 4t - 12 = 0$ **[M1]**

$t = 2$, or $t = -1.2$ (or $-\frac{6}{5}$) **[A1]**

[Back to Question] *[Total Marks: 7]*